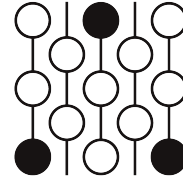


Review of Submission 10B

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

The content in this solution is not only incorrect on many levels, but also not mathematically salvageable. In a more basic sense, the proof does not even claim to address the question in hand. Below I list a collection of comments mostly pointing to errors:

1. In the section “problem solved”, the paper DKEP, which I assume stands for the paper of Ding-Kunnawalkam Elayavalli-Peterson titled “Properly proximal von Neumann algebras” is used in multiple locations without appropriate mathematical citations. Additionally several of the jargon used in the body are contrived and do not mean anything rigorous whatsoever (or at the very least, lack any rigorous definition). Examples include: “positive order of $\mathcal{N}$ ”, “support projection”, “normal bidual compactification”.
2. The formally established notation in the field, coming from the paper DKEP is not used at all in the body of this solution. For instance, by $\mathcal{N}$, it is possible that they mean $M^{\sharp*}$? Without establishing such a basic notation it is very hard to continue reading the proof.
3. The solution does not even explain what is the boundary piece X_U . It is meaningless because U is just defined using abstract sets, and there is no possible way to associate a boundary piece to this information.
4. Section 1, namely “A state-support fact” is a complete triviality that is standard in the field. So is Section 2, for that matter. These arguments in no way are non-trivial steps involved in the proof of the theorem at hand, therefore it appears that the author of the solution does not place an appropriate mathematical weight on the steps.
5. Ultimately, if one tries to be generous and extract the mathematics from this writeup, one only obtains a rather trivial fact about obtaining proper proximality combining relative proper proximality with respect to various boundary pieces whose supports intersect in the compact operators. This does not in any way constitute the technical difficulties in establishing the problem statement.

Correctness. The solution presented here does not address the main problem in any way, and hence is completely incorrect.

Novelty. The solution has no novelty. Even for the basic facts that it claims to prove such as the fact that $\phi(pap) = \phi(a)$ whenever ϕ is a state on a C^* -algebra that satisfies $\phi(p) = 1$, are extremely elementary and well known facts.

2 Recommendation

Should be rejected. This write up fails to meet even the most basic standards of rigor, and additionally does not answer the question at hand.

3 Submission with inline comments

Original Problem

Prove the following statement:

Let Γ be a finite simplicial graph with $|V(G)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in V$. If Γ is irreducible (in the sense that there does not exists subgraphs G_1, G_2 such that $V(G_1) \cup V(G_2) = V(G)$ and every $(v, u) \in E(G)$ for all $v \in V(G_1), u \in V(G_2)$), and each M_v has a trace zero unitary, then the graph product von Neumann algebra M is properly proximal.

Problem Solved

This write-up proves the following **abstract DKEP finite-cover upgrade lemma**, not the full graph-product theorem stated above. Thus it is a relaxation/different result: it assumes the relevant DKEP boundary pieces, support projections, support sandwich inequality, and relative proper proximality as hypotheses, and proves only the final upgrade from relative proper proximality for finitely many pieces to absolute proper proximality.

Let (M, τ) be a separable tracial von Neumann algebra, let $H = L^2(M, \tau)$, and let J be the modular conjugation, so that $\rho(x) = Jx^*J$ denotes the right action of M on H . Assume M is placed in the DKEP properly proximal framework: there is a unital von Neumann algebra $\mathcal{N}$, called the normal-bidual compactification, containing the left copy of M and containing support projections for DKEP boundary pieces. A DKEP boundary piece $X \subset B(H)$ means a hereditary C^* -subalgebra, invariant under the left and right M -actions, whose elements are treated as X -compact operators. Its DKEP support projection $q_X \in \mathcal{N}$ is the projection obtained as the weak-* limit in $\mathcal{N}$ of any contractive approximate identity of X , equivalently the least projection supporting the weak-* closure of X . The compact support $q_K \in \mathcal{N}$ is the support projection of $K(H)$.

Let $n \geq 1$, let $U_1, \dots, U_n$ be sets with finite-cover condition

$$\bigcap_{i=1}^n U_i = \emptyset,$$

and let

$$\mathcal{F} = \left\{ \bigcap_{i \in I} U_i : \emptyset \neq I \subseteq \{1, \dots, n\} \right\} \cup \{\emptyset\}.$$

For every $U \in \mathcal{F}$, suppose there is a DKEP boundary piece X_U with support projection $q_U \in \mathcal{N}$. Assume $q_\emptyset = q_K$. Assume also the support sandwich inequality

$$q_U q_W q_U \leq q_{U \cap W}$$

in the positive order of $\mathcal{N}$, for all $U, W \in \mathcal{F}$.

For a boundary piece X , define

$$S_X(M) = \{T \in B(H) : [T, \rho(x)] \in X \text{ for every } x \in M\}.$$

A state ψ on $S_X(M)$ is M -central if $\psi(xT) = \psi(Tx)$ for all $x \in M, T \in S_X(M)$. The algebra M is properly proximal relative to X if there is no M -central state on $S_X(M)$ whose restriction to the left copy of M is normal. The algebra M has no amenable direct summand if there is no nonzero central projection $z \in Z(M)$ such that zM is amenable.

Use the following DKEP obstruction theorem as a black box: if M has no amenable direct summand and is not properly proximal, then there exists an M -central obstruction state Ω on $\mathcal{N}$, normal on M , such that $\Omega(q_K) = 0$; moreover, if M is properly proximal relative to X_U , then every such obstruction state satisfies $\Omega(q_U) = 1$.

Under these assumptions, if M has no amenable direct summand and is properly proximal relative to each X_{U_i} , then M is properly proximal.

Proof

All products and inequalities below are taken inside the normal-bidual compactification $\mathcal{N}$. We deliberately do **not** assume that the projections q_U commute.

1. A state-support fact

Let A be a unital C^* -algebra, let ϕ be a state on A , and let $p \in A$ be a projection with $\phi(p) = 1$. Then

$$\phi(a) = \phi(pap) \quad \text{for every } a \in A.$$

Indeed, let $e = 1 - p$. Then e is a projection and

$$\phi(e) = \phi(1) - \phi(p) = 0.$$

By Cauchy–Schwarz for the positive functional ϕ ,

$$|\phi(ea)|^2 \leq \phi(e^*e)\phi(a^*a) = \phi(e)\phi(a^*a) = 0,$$

so $\phi(ea) = 0$ for all a . Similarly,

$$\phi(ae) = \overline{\phi(ea^*)} = 0.$$

Decomposing

$$a = pap + pa e + eap + eae,$$

and applying ϕ , all terms except pap vanish. Hence $\phi(a) = \phi(pap)$.

2. Two support-one projections force support one on the intersection

Let ϕ be a state on $\mathcal{N}$. Suppose $U, W \in \mathcal{F}$ and

$$\phi(q_U) = 1, \quad \phi(q_W) = 1.$$

Set

$$p = q_U, \quad s = q_W, \quad r = q_{U \cap W}.$$

By the state-support fact applied to p and $a = s$,

$$\phi(s) = \phi(psp).$$

Since $\phi(s) = 1$, we have

$$\phi(q_U q_W q_U) = 1.$$

By the support sandwich inequality,

$$q_U q_W q_U \leq q_{U \cap W} = r.$$

Because r is a projection, $0 \leq r \leq 1$, so

$$1 = \phi(q_U q_W q_U) \leq \phi(r) \leq 1.$$

Therefore

$$\phi(q_{U \cap W}) = 1.$$

No commutativity was used: $q_U q_W q_U$ is positive because

$$q_U q_W q_U = (q_W q_U)^*(q_W q_U),$$

and the argument uses only the compression $q_U q_W q_U$, not the product $q_U q_W$.

3. Apply the obstruction theorem

Assume, toward contradiction, that M is not properly proximal. Since M has no amenable direct summand, the DKEP obstruction theorem gives an M -central obstruction state Ω on $\mathcal{N}$, normal on M , such that

$$\Omega(q_K) = 0.$$

For each $i = 1, \dots, n$, M is properly proximal relative to X_{U_i} . Hence the second part of the obstruction theorem gives

$$\Omega(q_{U_i}) = 1 \quad \text{for every } i = 1, \dots, n.$$

4. Iterate the two-support argument

Define

$$V_k = U_1 \cap \cdots \cap U_k \quad (1 \leq k \leq n).$$

We prove by induction that

$$\Omega(q_{V_k}) = 1 \quad \text{for every } k.$$

For $k = 1$, this is exactly $\Omega(q_{U_1}) = 1$.

Assume $\Omega(q_{V_k}) = 1$. Since $\Omega(q_{U_{k+1}}) = 1$, the result of Section 2, applied with $U = V_k$ and $W = U_{k+1}$, gives

$$\Omega(q_{V_k \cap U_{k+1}}) = 1.$$

But

$$V_k \cap U_{k+1} = V_{k+1}.$$

Thus

$$\Omega(q_{V_{k+1}}) = 1.$$

By induction,

$$\Omega(q_{V_n}) = 1.$$

The finite-cover condition says

$$V_n = U_1 \cap \cdots \cap U_n = \emptyset.$$

Therefore

$$\Omega(q_{\emptyset}) = 1.$$

By hypothesis,

$$q_{\emptyset} = q_K.$$

Hence

$$\Omega(q_K) = 1.$$

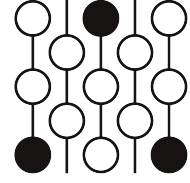
This contradicts the obstruction-state condition $\Omega(q_K) = 0$. Therefore the assumption that M is not properly proximal is false. Consequently, M is properly proximal.

Review of Submission 10B

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

This generated proof did not solve the original problem. Instead, it claims to prove a 'finite-cover upgrade lemma', which is in some sense in the same spirit of the original problem. In the proof of this 'upgrade lemma', there are many unjustified or wrong statements.

Correctness. In the setup of this lemma, there are several unclear notions: 1. The normal-bidual compactification is not defined explicitly, and this is not a standard terminology. Perhaps it means $\mathcal{N} := B(L^2M)_J^{\sharp*}$ or $\tilde{S}_X(M)$ in DKEP but it is not clear. 2. A DKEP boundary piece is not necessarily invariant under the left and right M -actions, it is only assumed to be invariant under the left and right action of a ultraweak dense C^* -subalgebra of M . 3. The definition of small-at-infinity boundary piece $S_X(M)$ is incorrect, it should be

$$S_X(M) = \{T \in B(H) : [T, \rho(x)] \in \overline{X}^{\infty,1} \text{ for every } x \in M\},$$

where $\overline{X}^{\infty,1}$ is the closure of X in $B(L^2M)$ with respect to the L^∞ - L^1 -norm. 4. The 'finite-cover condition' is not understandable. 5. The support sandwich inequality $q_U q_W q_U \leq q_{U \cap M}$ in fact implies that $[q_U, q_W] = 0$. So assuming this is the same as assuming $[q_U, q_W] = 0$, which is the key difficulty of the proof.

Novelty. The presented version seems to reduce the original problem to a seemingly weaker theorem by assuming several conditions. But in fact, the assumptions are the most difficult parts. Mathematically speaking, there are not many new ideas involved. The proof of section 2 has some novelty, but this part can be completely avoid by directly proving $[q_U, q_W] = 0$ from the assumption.

2 Recommendation

Should be rejected as it fails to solve the problem.

3 Submission with inline comments

Original Problem

Prove the following statement:

Let Γ be a finite simplicial graph with $|V(G)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in V$. If Γ is irreducible (in the sense that there does not exist subgraphs G_1, G_2 such that $V(G_1) \cup V(G_2) = V(G)$ and every $(v, u) \in E(G)$ for all $v \in V(G_1), u \in V(G_2)$), and each M_v has a trace zero unitary, then the graph product von Neumann algebra M is properly proximal.

Problem Solved

This write-up proves the following **abstract DKEP finite-cover upgrade lemma**, not the full graph-product theorem stated above. Thus it is a relaxation/different result: it assumes the relevant DKEP boundary pieces, support projections, support sandwich inequality, and relative proper proximality as hypotheses, and proves only the final upgrade from relative proper proximality for finitely many pieces to absolute proper proximality.

Let (M, τ) be a separable tracial von Neumann algebra, let $H = L^2(M, \tau)$, and let J be the modular conjugation, so that $\rho(x) = Jx^*J$ denotes the right action of M on H . Assume M is placed in the DKEP properly proximal framework: there is a unital von Neumann algebra $\mathcal{N}$, called the normal-bidual compactification, containing the left copy of M and containing support projections for DKEP boundary pieces. A DKEP boundary piece $X \subset B(H)$ means a hereditary C^* -subalgebra, invariant under the left and right M -actions, whose elements are treated as X -compact operators. Its DKEP support projection $q_X \in \mathcal{N}$ is the projection obtained as the weak-* limit in $\mathcal{N}$ of any contractive approximate identity of X , equivalently the least projection supporting the weak-* closure of X . The compact support $q_K \in \mathcal{N}$ is the support projection of $K(H)$.

Reviewer 2

The definition of $\mathcal{N}$ is not clear and the 'normal-bidual compactification' terminology is not standard.

Let $n \geq 1$, let $U_1, \dots, U_n$ be sets with finite-cover condition

$$\bigcap_{i=1}^n U_i = \emptyset,$$

and let

$$\mathcal{F} = \left\{ \bigcap_{i \in I} U_i : \emptyset \neq I \subseteq \{1, \dots, n\} \right\} \cup \{\emptyset\}.$$

For every $U \in \mathcal{F}$, suppose there is a DKEP boundary piece X_U with support projection $q_U \in \mathcal{N}$. Assume $q_\emptyset = q_K$. Assume also the support sandwich inequality

$$q_U q_W q_U \leq q_{U \cap W}$$

in the positive order of $\mathcal{N}$, for all $U, W \in \mathcal{F}$.

This support sandwich inequality in fact implies that $[q_U, q_W] = 0$, which is the main difficulty of the problem.

For a boundary piece X , define

$$S_X(M) = \{T \in B(H) : [T, \rho(x)] \in X \text{ for every } x \in M\}.$$

A state ψ on $S_X(M)$ is M -central if $\psi(xT) = \psi(Tx)$ for all $x \in M, T \in S_X(M)$. The algebra M is properly proximal relative to X if there is no M -central state on $S_X(M)$ whose restriction to the left copy of M is normal. The algebra M has no amenable direct summand if there is no nonzero central projection $z \in Z(M)$ such that zM is amenable.

Use the following DKEP obstruction theorem as a black box: if M has no amenable direct summand and is not properly proximal, then there exists an M -central obstruction state Ω on $\mathcal{N}$, normal on M , such that $\Omega(q_K) = 0$; moreover, if M is properly proximal relative to X_U , then every such obstruction state satisfies $\Omega(q_U) = 1$.

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$$\phi(ae) = \overline{\phi(ea^*)} = 0.$$

Decomposing

$$a = pap + pae + eap + eae,$$

and applying ϕ , all terms except pap vanish. Hence $\phi(a) = \phi(pap)$.

This section is basic knowledge, and could be skipped.

2. Two support-one projections force support one on the intersection

Let ϕ be a state on $\mathcal{N}$. Suppose $U, W \in \mathcal{F}$ and

$$\phi(q_U) = 1, \quad \phi(q_W) = 1.$$

Set

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No commutativity was used: $q_U q_W q_U$ is positive because

$$q_U q_W q_U = (q_W q_U)^*(q_W q_U),$$

and the argument uses only the compression $q_U q_W q_U$, not the product $q_U q_W$.

This part can be simplified into one line by realizing that the assumption implies $[q_U, q_W] = 0$.

3. Apply the obstruction theorem

Assume, toward contradiction, that M is not properly proximal. Since M has no amenable direct summand, the DKEP obstruction theorem gives an M -central obstruction state Ω on $\mathcal{N}$, normal on M , such that

$$\Omega(q_K) = 0.$$

For each $i = 1, \dots, n$, M is properly proximal relative to X_{U_i} . Hence the second part of the obstruction theorem gives

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$$\Omega(q_{\emptyset}) = 1.$$

By hypothesis,

$$q_{\emptyset} = q_K.$$

Hence

$$\Omega(q_K) = 1.$$

This contradicts the obstruction-state condition $\Omega(q_K) = 0$. Therefore the assumption that M is not properly proximal is false. Consequently, M is properly proximal.